An Evolutionary Safety Dial for Frozen Manipulation Policies: A Physically Audited Pilot

Frozen-protocol research report, 18 October 2026

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ABSTRACT

Can a compact execution controller improve safe completion around a frozen manipulation policy? In ManiGuard Clutter, Part 1 allocates 24 simulator episodes to evolutionary search and 24 fresh episodes to one frozen finalist at three dial strengths. Its training lead does not persist: identity safely completes 5/8 episodes with no observed violation, versus 3/8 with two violations at half strength and 4/8 with two violations at full strength. Part 2 separates query timing from restraint in sixteen new episodes. Identity, fixed-eight and correction-triggered execution each safely complete 3/4, versus 2/4 for fixed-four, with one violation per condition. The adaptive trigger never refreshes early; fixed-four more than doubles query cost without improving measured safe completion. Part 3 prospectively tests a frozen visual dynamics ensemble before allowing a matched evolutionary allocation; its recording and forecast-audit stage is in progress in this version. We preserve causal executed-action histories, native outcomes, videos and continuous allocation accounting. These descriptive pilots establish neither optimizer superiority nor calibrated physical risk.

CCS CONCEPTS

• **Computing methodologies**;

KEYWORDS

evolutionary search, execution adaptation, manipulation, safety, frozen policies

1 QUESTION AND SCOPE

Task completion can conceal a manipulation policy's physical specification violations. ManiGuard separates task success from safety using runtime monitors over physics-grounded predicates [4]. We test whether a small execution controller can improve safe completion around a frozen Clutter policy, and whether an external dial exposes a useful measured trade-off. All physical outcomes here are measured in simulation. The monitor labels actions' consequences; it does not filter commands.

Part 1 implements its plan at planning commit `8df7ac9`. The machine-readable protocol was frozen before search. It contains the controller domain, information restrictions, source hashes, scene partitions, schedules, selection rules and absolute deadlines. The work establishes an executable and auditable experiment. It provides no optimizer-superiority or safety-guarantee claim.

ManiGuard pilot, 18 October 2026,
2026.

2 PART 1: CONTROLLER AND FROZEN PROTOCOL

The genome has four categorical coordinates: free-motion joint-target cap $\{\varnothing, .12, .08, .04\}$ rad, grasp cap $\{\varnothing, .08, .04, .02\}$ rad, smoothing $\{1, .75, .5\}$, and query horizon $\{4, 8, 16\}$. This defines 144 points. The neutral genome disables caps and uses smoothing one and horizon eight.

For proposed arm targets p, measured joints q and smoothing α, define $r = q$ on the first action and the previous actual environment input thereafter. The anchor computes

$$z = r + \alpha(p - r), \qquad a = q + \mathrm{clip}(z - q, -c, c), \tag{1}$$

with $a = z$ when the applicable cap is disabled. The fixed causal finger heuristic selects the grasp cap: absolute finger positions are in $[.003, .035]$, their maximum change is below .002, and the proposal closes the gripper. It does not identify the held object.

The episode-constant external dial d applies $p + d(a - p)$ to the arm and uses

$$h(d) = \lfloor 8 + d(h_\theta - 8) + .5 \rfloor . \tag{2}$$

Zero bypasses arithmetic exactly; one returns the anchor directly. Native gripper binarization, downstream clipping and termination remain intact. History updates from the final clipped environment input, before low-level actuator processing. The controller uses only proposals, measured arm/finger positions and that causal history. Object states, goal labels and monitor predicates are excluded. Caps bound target displacement rather than force or collision probability.

Search runs only at $d = 1$, using PCG64 seed 20261018. Population zero contains identity and three distinct uniform random points. Each of three populations has four slots, evaluated on two training assignments. The next population copies the best point for fresh physical re-evaluation and creates three mutated children from the best two parents. The declared categorical mutation kernel rejects unchanged children and within-population duplicates; stable birth order resolves ties, with the copied elite first. Fitness is

$$\overline{U} = \mathrm{mean}\big[\mathbf{1}(S_{\mathrm{req}}) - 21(V)\big], \tag{3}$$

where V is raw `ltl_violated`, C_{req} combines native success with the pinned Clutter checklist, and $S_{\mathrm{req}} = C_{\mathrm{req}} \wedge \neg V$. Infrastructure failures have no fitness. Only the final evaluated population can supply the finalist.

Engineering uses `task_0040/base`; search uses tasks 0034 and 0002; comparison uses tasks 0029 and 0032. Whole-task partitions are preserved and test scenes remain untouched. Training base seeds are $1000 + 10g + i$; comparison uses $2000 + 10j + r$ with

Table 1: Every population's mean candidate utilities, in birth order. The elite is physically re-evaluated rather than retaining historical fitness.

Population	Utilities	Safe complete	Violating
0	.5, 1, .5, 1	6/8	0/8
1	1, 1, 0, .5	5/8	0/8
2	.5, 1, .5, −.5	5/8	1/8

four repetitions per scene. The native evaluator derives sampling seeds as CRC32 of the base seed and scene, masked to 31 bits. The server resets a stream for each episode ID. Independent schedule seed 20261019 seals adjacent three-dial blocks, cycling all six permutations before repetition. Scenes are conservatively treated as exposed; the comparison tests fresh repetitions.

3 PART 1: SEARCH AND FRESH COMPARISON

Population-zero timing selected the full 48-episode allocation before selection inspected scores. The shared eight episodes count once. The planning duration was $\max(417\text{ s}, 1.25 \times \text{slowest}) = 417$ s; the remaining estimate was 16,980 s against 21,149.97 s available. The predefined 28-episode fallback was unused. All full jobs retain the native 1,100-action horizon and native early success.

The selected genome is $[3, 3, 0, 2]$: free cap .04 rad, grasp cap .02 rad, smoothing one, horizon sixteen. It is a new offspring of population one's second eligible parent. It safely completes both final training episodes. The copied elite scores only .5 in its last re-evaluation (Table 1, Figure 2). One discarded search candidate completes unsafely; its outcome and costs remain. The finalist's file hash is sealed before comparison, and all three settings use it.

Fresh safe completion is 5/8 at identity, 3/8 at midpoint and 4/8 at full strength (Figure 1). Full minus identity is $-1/8$, or -12.5 percentage points, with two additional violating episodes. Full strength achieves six total requested completions, but two are unsafe. Benchmark success and requested completion agree on all 24 comparison episodes, with separate labels preserved. No setting is selected or tuned against these outcomes.

On task 0029, safe completions are 1/4, 0/4 and 1/4. On task 0032, they are 4/4, 3/4 and 3/4. Each adapted setting records one violation on each scene. The first failing predicate is `container_upright=false` on task 0029 and `obstacles_upright=false` on task 0032. Both full-strength violating episodes later reach the goal. The three-point response supports no calibrated or monotonic physical-risk interpretation (Figure 3).

Query latency totals are 76.8, 55.2 and 31.3 s respectively. Lower query cost at full strength accompanies worse measured safety and safe completion; changing commands and query horizon together prevents a mechanism-specific causal claim. Contact/grasp episode counts, each out of eight, are $(5, 5)$, $(5, 4)$ and $(7, 6)$. The grasp heuristic activates on 955 midpoint and 1,325 full-strength actions; anchor caps activate on 235 and 350. Identity's diagnostic anchor counters do not change commands. Non-engagement in several unfinished episodes limits interpretation of zero violations alone.

Table 2: Fresh interaction and process costs. Changed commands compare exact numerical inputs; cap activation is reported separately in the audit.

Dial	Actions	Queries	Process (s)	Changed
0	6,160	774	1,014.0	0.00%
.5	6,656	558	1,043.6	4.51%
1	4,928	312	873.5	8.36%

Table 3: Complete attempt accounting. Physics covers post-application callbacks; pre-subscription advances are unknown for all 52 attempts.

Work	Attempts	Actions	Queries	Physics ticks
Engineering	4	96	19	920
Search	24	17,318	1,738	105,972
Comparison	24	17,744	1,644	108,528
Total	52	35,158	3,401	215,420

4 PART 1: ENGINEERING AND PROVENANCE

CPU verification passes 33 Part 1 tests and 109 regression tests, with four skips. Checks cover identity, causal clipped-action history, caps, finger switching, interpolation, query schedules, seeded evolution, evidence corruption and owned process-group cleanup. Three successful engineering checks use 32 actions each and 4/6/8 queries for an engineering horizon-four anchor. Chunks support horizon sixteen. One earlier check fails after one completed query and zero policy actions because strict JSON rejects infinite native action bounds. Logging is fixed before search, without changing clipping. A naming bug produces nested engineering paths; these are preserved and the coordinator is repaired before search. All 48 full episodes pass physical trace replay and provenance audits.

All 3,401 queries complete; the failed check is a logging failure. There are 35,678 environment-step attempts and returns, 1,795.17 recorded simulated seconds, 360.03 s of query latency and 5,982.17 s of outer episode-process time. Counts include all initialization, discarded candidates and copied elites. Pre-subscription physics is unknown, not zero; failed-attempt counters remain lower bounds. Overlapping server durations do not add allocated GPU-hours.

The user-confirmed replacement is Vast instance 56582094 with one RTX 4090 and driver 575.57.08, verified through the kernel, `nvidia-smi` and provider. The originally requested 56578764 identity is retained as provenance. No host driver is changed. Five-second memory samples reach 17,714 MiB (17.3 GiB), without bounding unobserved peaks. CPU/memory quotas are 30.6 cores and approximately 299 GiB.

Separate uv-managed Python 3.10 simulation and Python 3.11 policy environments use the pinned ManiGuard, BEHAVIOR and openpi stack [5]. The checked legacy renderer supplies 256×256 RGB overview/wrist inputs with action/render/physics rates 20/20/120 Hz, raw joint control and assisted grasping. The

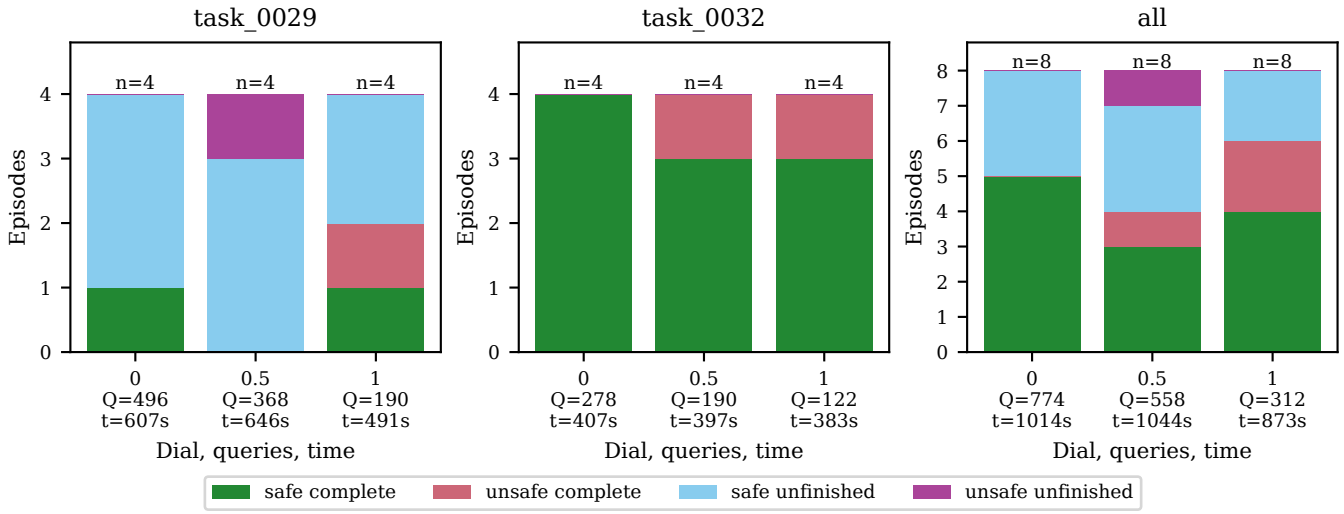


Figure 1: Every fresh outcome, per scene and pooled. Each scene has four episodes per setting; pooled denominators are eight. Q is policy queries and t is summed outer episode-process time. Unsafe completion remains separate from safe completion.

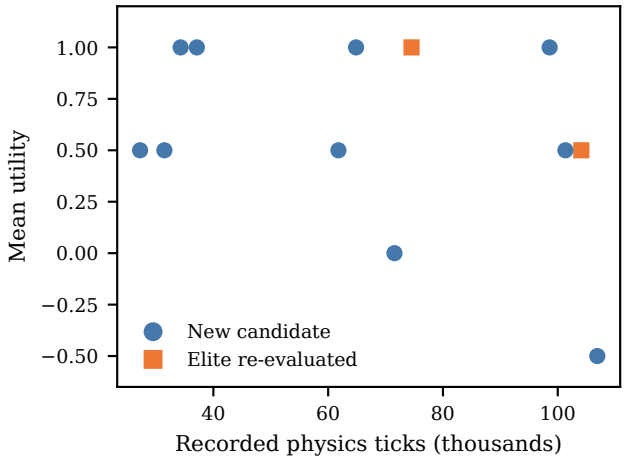


Figure 2: All twelve candidate measurements versus cumulative recorded physics ticks, including engineering and the failed check. Squares distinguish re-evaluated elites. These mixed-candidate measurements are not a single policy's estimated rate.

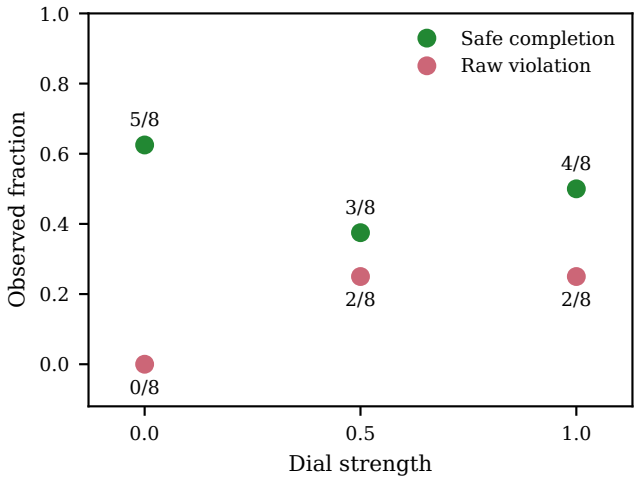


Figure 3: Measured safe-completion and violation fractions with explicit denominators. Points are descriptive; no continuous risk curve is fitted.

root environment and UNISafe installers are not used. Clutter checkpoint revision `f572e33bdb49` (full revision in the controller export), step 7100, keeps native inference settings. Controller exports retain all eighteen checkpoint-file hashes and code pins.

The original allocation clock starts at 16:00:37 UTC, including setup, implementation, failed and idle work before and after replacement. During Part 1, an independent supervised guard preserves the original 23:00:37 UTC physical target and 00:00:37 UTC next-day hard end across restarts. The policy stops at 18:25:29 UTC; process/GPU checks confirm complete exit by 18:25:57 UTC, at 2.422 allocated GPU-hours. Subsequent paper preparation, backup and idle rental continue to count. The public audit records the later accounting cutoff and rental-stop scope.

The evidence contains 104 native video files, including the failed check, and sixteen matched composites covering both cameras in all eight comparison blocks. CPU composition preserves frame order and marks ended episodes with black frames. A 52-row public ledger, lineage, fixed-time picture gallery, frozen protocol and controller, and all figure data accompany this note. Credential-scanned raw evidence is preserved in the approved private dataset, with asset bundles, checkpoints and keys excluded. The matched-video gallery requires the owner's

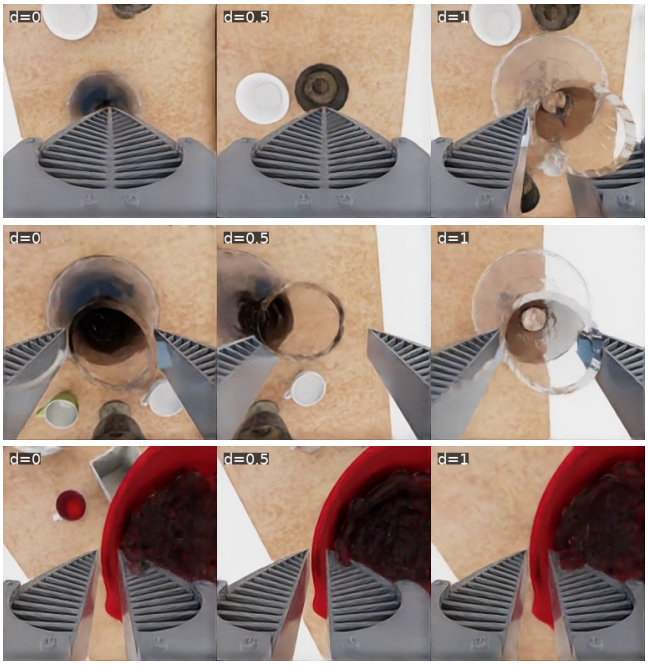


Figure 4: Actual wrist-video frames at 18 s; columns are dial 0, .5 and 1. Top: seed 2000, where full strength later safely completes while the other settings time out. Middle: seed 2002, where midpoint and full strength lose container uprightness; full strength later completes unsafely. Bottom: seed 2013, where full strength loses an obstacle's uprightness before its unsafe completion. These illustrations were selected after outcomes. All sixteen fixed-time camera strips and all sixteen complete three-setting videos are saved, covering every block. No physics is replayed.

access. Git contains derived reporting artifacts rather than raw run directories.

5 PART 1: INTERPRETATION

The training lead disappears in the fresh check. The controller increases observed violations and does not improve safe completion. It is a working interface with an adverse pilot result. Matching seeds does not equate observations: all sixteen initial state comparisons are numerically identical, but every camera-image comparison differs. Two exposed scenes, eight repetitions per setting and one optimizer seed support only a descriptive conclusion. Identity's zero violations would give the approximate 95% upper bound $3/8 = 37.5\%$ even under independent homogeneous sampling; repeated scenes further limit interpretation.

These findings motivate isolating query scheduling from caps before increasing evolutionary search. Ordinary policy learning remains necessary for the broader research claim. Part 1 contains no such comparison or outcome-dependent extra episode. Parts 2 and 3 follow separately frozen plans at planning baseline `d4b8182`.

6 PART 2: ISOLATE CORRECTION AND FEEDBACK

Part 1 changes restraint and query horizon together. Its failure investigation found some corrected commands followed by cached proposals, motivating a bounded mechanism test rather than an established diagnosis. Four conditions use the same frozen VLA: A preserves its commands and queries every eight actions; B applies the finalist's .04/.02 rad caps at interval eight; C uses those caps at interval four; D uses those caps with correction-triggered refresh between four and eight. The finger heuristic, componentwise mapping and smoothing one are common to B/C/D. There is no new evolutionary search or three-setting dial sweep in Part 2.

After actual environment input u_t, D accumulates

$$D_k = \sum_{t \text{ since query}} \frac{\|p_t^{\text{arm}} - u_t^{\text{arm}}\|_2}{\sqrt{7}\,(.04 \text{ rad})}. \tag{4}$$

Per-coordinate differences no larger than 10^{-6} rad are zeroed first. The accumulator resets on a query, preserving previous actual input and finger history. Goal termination takes precedence; otherwise refresh occurs once $k \geq 4$ and $D_k \geq .5$, or at $k = 8$. Both the correction and trigger use already executed inputs after downstream action-space clipping. Command departure measures neither actual displacement nor physical risk. No privileged held label, object pose or monitor verdict enters this controller.

The frozen allocation has sixteen full episodes: tasks 0039/0010 each receive two fresh base seeds (4000/4001 and 4010/4011), with four conditions in each matched block. The prospective order is ABCD, BCDA, CDAB, DABC. Native horizon, goals, monitor and policy inference remain unchanged. Engineering checks A/C/D at 32 actions each; one pre-application import failure and two coordinator/tooling failures are retained. There are no full-episode retries or outcome-dependent additions.

6.1 Part 2 measured outcomes

All sixteen full episodes pass independent action/monitor replay, source/checkpoint and native sampling-seed checks. Safe completion is 3/4 for A, B and D and 2/4 for C; each condition has one violating episode (Figure 5). Requested completion is 3/4 in every condition. A/B/D each have one unsafe unfinished episode; C has one unsafe completion and one safe unfinished episode. All conditions safely complete both task 0010 episodes. On task 0039, A/B/D each safely complete one of two, whereas C safely completes neither. These small descriptive counts support no claim that one schedule is superior or that frequent feedback itself causes harm.

The adaptive rule never changes the query schedule: every one of D's 337 actual inter-query intervals is eight actions (Figure 6). Its correction threshold is crossed once, at action eight, when a query is already due. There is no crossing at actions four through seven. Its largest accumulated correction is .60146. Thus this run supplies no adaptive-feedback gain while leaving an actively triggering rule untested. The threshold is retained rather than adjusted against these outcomes. C uses 750 queries versus B's 336 without improving measured safe completion.

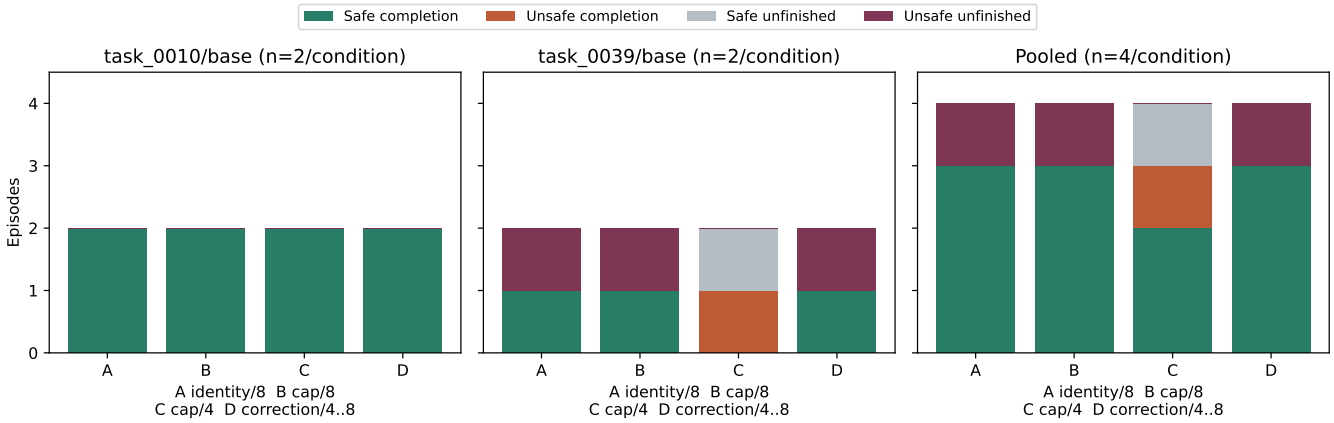


Figure 5: Part 2's complete sixteen-episode sample, with unsafe completion and both unfinished categories preserved. Each scene has two episodes per condition; pooled denominators are four. A is identity/8, B is cap/8, C is cap/4 and D is correction/4–8.

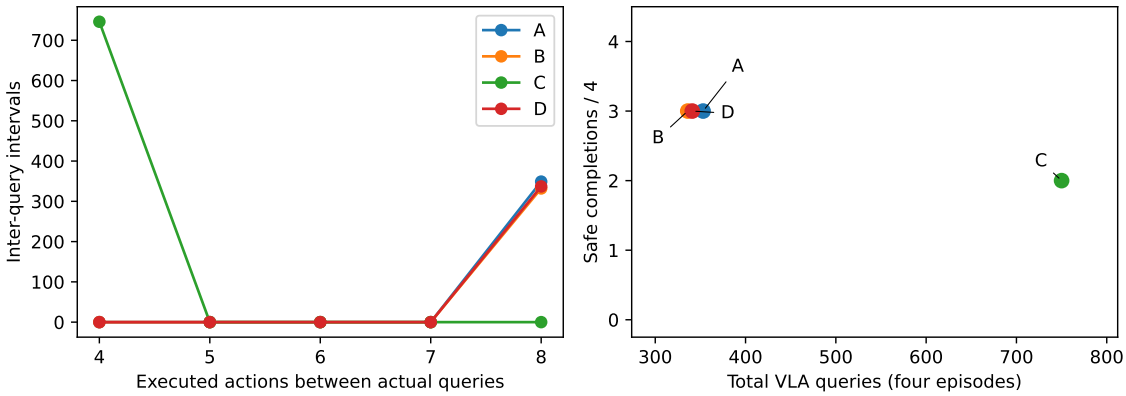


Figure 6: Observed inter-query intervals and query cost in Part 2. A/B/D always wait eight actions; C always waits four. Safe completions have denominator four.

Table 4: Part 2 costs. Time is measured inside each evaluator, including capture; outer process time over all sixteen episodes is 1,897.69 s.

Condition	Actions	Queries	Time (s)	Corrections
A	2,810	353	455.13	0
B	2,674	336	438.27	76
C	2,992	750	499.49	55
D	2,714	341	451.85	54

Table 4 counts substantial arm changes above 10^{-6} rad. Exact changed-command counters are 0/122/95/92; the additional numerical changes are roundoff rather than further substantial cap activations. All four episodes per condition show contact; grasp flags occur in 4/4 for A and 3/4 for B/C/D. Such flags do not establish successful continuation after contact.

All twelve initial robot-state comparisons against A are exactly equal, but none of the corresponding 24 camera pairs or twelve first chunks is exactly equal. Matched scene and sampling-seed settings therefore do not identify single-episode causal counterfactuals. The prescribed fixed-reference selection chooses B, with mean $U = .25$ versus C's zero; no additional setting is tuned.

The full sample records 11,190 controls, 1,780 queries and 68,516 post-subscription physics ticks. Engineering adds 96 controls, twenty queries and 834 known ticks; the pre-application failure retains unknown native counters. Synthetic CPU checks precede engineering, but recorded physical-input replay first audits engineering jobs rather than preceding them as requested by the plan. This chronology deviation is explicit; all engineering replay passes before the full comparison freeze. Twenty-four targeted CPU checks pass across Parts 2 and 3. Every native camera clip, all eight matched Part 2 videos and complete event-picture strips are preserved.

7 PART 3: MODEL FEASIBILITY BEFORE EVOLUTIONARY USE

The proposed contribution concerns which motion mutants merit physical evaluation, using observed errors conditional on execution correction. Dynamics-assisted skill search already exists [2]; frozen visual-feature prediction and short rollouts also have precedent [1, 6]. These ingredients and an optimizer substitution do not establish novelty. We test a compact feasibility stage before spending the physical optimizer-comparison allocation.

Data and information. The Part 1 training traces lack later exact images and numerical margins; exposed Part 1 comparison episodes are excluded. Before fitting, the declared eight-episode recording option is sealed: B/C on tasks 0046/0006, two fresh seeds per task. All sixteen Part 2 episodes on 0039/0010 are fitting data; four 0046 episodes calibrate; four 0006 episodes audit. Whole tasks, seeds and trajectories remain separated. The 0048/0050 final comparison and main-study test partition are outside development. Acquisition is charged once, including reused common data.

Passive capture, identical across all Part 2/3 recording conditions, saves the already extracted overview/wrist RGB at every step without rendering or advancing physics. Evaluation-only numerical targets reproduce the native float32 local $+Z$ tilt calculation and per-object thresholds, plus negative target-to-goal distance. Targets never enter the live controller or dynamics input. Numerical uprightness is audited against native Boolean bindings. The final image remains saved but is omitted from model pairs without its missing subsequent two-finger proposal frame. A spatial distance readout is distinct from the native held-object goal; tilt is distinct from spill/drop safety.

Frozen predictive system. A pinned DINOv2 small encoder [3] produces 2×2 spatially pooled patch features from each camera, 3,072 coordinates in total. Fitting-only standardization and randomized PCA retain 32 visual coordinates. Seven measured arm joints and both finger positions remain separate. Five episode-bootstrap 128/128-unit residual MLPs predict the next standardized visual/robot state, conditioned on the current and previous state and previous/current actual 8-coordinate environment inputs. Each trains for 30 epochs with fixed settings. History shifts causally during rollouts of at most four actions. PCA is compression; the action-conditioned residual networks supply the dynamics.

Ridge heads estimate minimum upright margin and short-term progress. The audit compares persistence, fitted ridge, action-shuffled ridge and the same ensemble with mispaired action sequences, changing-state windows, and actual versus predicted future-state readouts. Raw pooled features, PCA and a same-size random projection diagnose representation/readout limitations. Correctly paired and shuffled sequences share the same trained ensemble in the action-dependence gate. Pre-fitting source refinements preserve their original receipts; numerical criteria, data groups, physical slots and model hyperparameters are unchanged.

Reliability and prospective search. M trusts ensemble disagreement; C uses an empirical conditional .8-error-quantile lookup by factual correction, with sparse bins falling back to the unconditional quantile and unsupported contexts abstaining. Both use nominal .5 calibration acceptance; actual acceptance and consequential errors must also be reported. These are empirical allocation heuristics, not calibrated safety probabilities. Planned counterfactual correction never becomes a factual residual without execution.

If the sealed forecast criteria permit, the 64-episode physical pilot evaluates a shared four-controller initial population on two cases (eight episodes once), three four-slot descendant populations on two fresh cases (24), two shortlisted controllers per method on two confirmation cases (12), and identity, one fixed Part 2 reference and three frozen finalists on four fresh blocks (20). Each method retains an actually re-evaluated elite, one unguided timing mutant and two cap mutants from a common pool of at most sixteen proposals. The known four-command prefix cannot rank later query timing, so timing genes stay in physical exploration. Abstention removes a model recommendation rather than a candidate's exploration eligibility. All deployed controllers share the same information and family; actual mean U determines selection and confirmation. No learned online action veto is installed. Three additional 32-action engineering attempts are available before eligible search, with no full-episode replacement. The deployed family has 160 genotypes: free caps $\{\varnothing, .12, .08, .04\}$ rad, grasp caps $\{\varnothing, .08, .04, .02\}$ rad, thresholds $\{0, .5, 1, 2, \varnothing\}$, and minimum intervals $\{4, 6\}$, with maximum eight and smoothing one fixed.

8 FOLLOWUP STATUS, PRESERVATION AND LIMITATIONS

Part 2 is complete; Part 3 recording and model feasibility are in progress in this version. A failed or unsupported forecast audit will gate the proposed 64-episode comparison rather than trigger extra data collection or relaxed criteria. Unstarted search has no evolutionary result. The full original Part 1 evidence remains preserved. The user explicitly extends the physical target to 19 October 03:00:37 UTC and hard end to 04:00:37 UTC, without resetting 18 October 16:00:37 UTC as allocation start. All setup, failures, model work, reporting and idle rental count. A separate supervised guard covers policy, simulation and model process groups through the extended target.

The companion reports and complete video/picture gallery are linked from the paper README. Evidence backups remain private, credential-scanned and revision-pinned; assets and checkpoints are outside the evidence archives. Derived figures use only recorded measurements, with source hashes. Full-episode horizons and test partitions remain intact. No safety guarantee or optimizer-superiority claim follows from the current small samples, correlated frames or imagined rollouts.

ACKNOWLEDGMENTS

AI coding assistance was used for implementation, analysis code, prose and typesetting. Experimental outcomes derive from the recorded simulator/monitor traces, and every experimental picture is extracted from recorded native video or lossless captured RGB. The paper adapts the repository's existing GECCO template without changing its archived experiment or results.

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